2026-06-10-SP2-helixagent-exposure-plan

SP2 — HelixAgent Exposure Extension — Implementation Plan

Field	Value
Revision	1
Last modified	2026-06-10
Status summary	PLANNING — TDD/anti-bluff plan; NOT approved for implementation (brainstorming HARD-GATE per master roadmap §0/§4)
Sub-program	SP2 (depends on SP1; feeds from analysis `…-C-…`)
Source design	`docs/superpowers/specs/analysis/2026-06-10-C-helixagent-exposure.md`
Roadmap	`docs/superpowers/specs/2026-06-10-llms-access-master-roadmap.md` §SP2
Scope root	`/Volumes/T7/Projects/HelixCode/submodules/helix_agent` (module `dev.helix.agent`, `go 1.26`) + `helix_code/go.mod` (SDK currency only)
Authority	constitution/ + root CLAUDE.md/ CONSTITUTION.md; anti-bluff §11.4 family governs every closure

> **Anti-bluff note (§11.4 / §11.4.123):** This is a *plan*, not a claim of done work. No task below is "done" until it carries captured runtime evidence per §11.4.5 / §11.4.69 / §11.4.107. Every fix is RED-first (§11.4.115): reproduce the defect on the pre-fix artifact with a `RED_MODE` polarity switch BEFORE writing the fix. Code-review gate (§11.4.125/§11.4.134) runs before pre-build + main build. Regression guards (§11.4.135) registered in the same commit as each fix.

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0. Goal (restated from roadmap §SP2)

Under **ONE root**, expose as uniformly-named, individually-addressable selection targets: 1. the **AI-debate ensemble** (aggregate + named presets), 2. **HelixLLM** promoted to a first-class root entry (when `USE_HELIX_LLM=true`), 3. **every discovered provider individually**, and 4. **each provider's WORKING (verified) models** (`DiscoveredModel.Verified == true`).

Plus: update provider/API SDKs to latest (deep-web-research §11.4.99); resolve the gin 1.11.0 ⍰ 1.12.0 module skew; ship docs/guides/graphs/SQL/website/video-course deliverables and the full test-type matrix.

Naming scheme (analysis-C, confirmed against existing conventions): ensemble · ensemble/<preset> · helixllm · helixllm/<model> · <provider> · <provider>/<model_id> (preserving already-namespaced ids like openrouter/x-ai/grok-4).

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1. Verified architecture grounding (re-confirmed against live source, 2026-06-10)

All re-opened and confirmed this session; line numbers are current HEAD (a few drifted ±5 lines from the analysis doc — corrected here):

Fact	Evidence (file:line)
Exposure surface = Gin routes, all under /v1	internal/router/router.go (route block from
Ensemble route (inline closure → GetEnsembleService().RunEnsemble)	internal/router/router.go:676 (call at :73 object tag "ensemble.completion" :740; defaul EnsembleConfig.Strategy="confidence_we
Providers list (reads live GetCapabilities().SupportedModels)	internal/router/router.go:773 (GET "" cl :773-797, models field :783)
Discovery group (models/selected/stats/trigger/ensemble/debate-model)	internal/router/router.go:1306-1313
Verification group (LLMsVerifier)	internal/router/router.go:1378-1387; bri importing live providers :1356-1379
Runtime registry services.ProviderRegistry	internal/services/provider_registry.go
HelixLLM = gated provider "helixllm" only (no root)	internal/services/provider_registry.go:750-764 (USE_HELIX_LLM :755, model id helixllm-default :757)
LLMProvider interface — **NO** GetModels()	internal/llm/provider.go:9-14 (only Comp CompleteStream/HealthCheck/GetCapabili ValidateConfig)
“Working model” flag (verifier authoritative)	internal/verifier/discovery.go:56 (DiscoveredModel.Verified; OverallScore ensemble member shape SelectedModel :66-72
Discovery service API	internal/verifier/discovery.go:23; GetSelectedModels() / GetDiscoveredMode
models.dev reference catalog (distinct from runtime lists)	internal/modelsdev/service.go:13-23, client.go:14
D-1 BLUFF: /v1/completion/models **hardcoded 3-model list**	internal/handlers/completion.go:411 (Mo func; literal maps :413-436: deepseek-coder, claude-3 sonnet-20240229, gemini-pro, time.Now() stamps)

Fact	Evidence (`file:line`)
helix_agent has **zero vendor SDKs** (48 hand-rolled `net/http` providers)	`submodules/helix_agent/go.mod` (no aws/azur anthropic/genai)
Cloud SDKs live in inner app	`helix_code/go.mod` (azcore v1.16.0 `:12`, azidentit `:13`, aws-sdk-go-v2 v1.32.7 `:15`, bedrockruntime v1.23 zep-go/v3 v3.10.0 `:26`, tree-sitter Aug-2024 pin `:39`, gi `:27`, pgx v5.7.6, go-redis v9.17.2)
gin module skew	`helix_code/go.mod` gin **v1.11.0** vs `submodules/helix_agent/go.mod` gin **v1.12.0** (line 5)

Design consequence: the unified-catalog work is a NEW handler + a join over three already-existing sources (`ProviderRegistry.ListProviders()` → `GetCapabilities().SupportedModels`, `verifier` discovery `Verified`/`OverallScore`, and the ensemble service). No source is removed (§11.4.122); all current endpoints stay intact (additive).

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2. Operator-input decisions (FLAGGED — resolve before implementation)

These are design forks the agent must NOT decide silently (§11.4.101 — reversible/low-blast items defaulted; high-blast items blocked for operator):

#	Decision	Default proposed (if operator silent)	Why it needs operator
OP-1	**Root path for the unified catalog.** `GET /v1/catalog` (analysis-C illustrative) vs `/v1/exposure` vs `/v1/targets`.	`GET /v1/catalog`	Public API surface naming; appears in docs/website/SDK; hard to rename later.
OP-2	**Add `GetModels(ctx) ([]Model, error)` to the `LLMProvider` interface** (T2.2.2) vs keep reading `GetCapabilities().SupportedModels` + verifier join only.	Keep interface unchanged for SP2; join via verifier (smaller blast radius — 48 providers would each need the new method).	Interface change touches all 48 providers + tests; CONST-036 single-source.
OP-3	**Ensemble preset enumeration source.** Expose only `ensemble` + `ensemble/confidence_weighted`, or enumerate every `EnsembleConfig.Strategy` value as a preset.	Expose `ensemble` (alias `ensemble/auto`) + presets derived from the strategies actually wired in code.	Avoids advertising presets that aren't implemented (would be a §11.4

#	Decision	Default proposed (if operator silent)	Why it needs operator
			capability bluff).
OP-4	**SDK update blast radius (P2.3).** Update ALL flagged SDKs to latest now, or only the high-value/security ones (bedrockruntime, azidentity) this cycle.	Update security/ auth + high-value (azidentity, azcore, bedrockruntime, aws-sdk-go-v2 core) this cycle; defer cosmetic minors with a tracked item.	Major-version bumps can break build; §11.4.92 multi-pass + §11.4.108 four-layer needed per bump.
OP-5	**gin skew resolution direction (P2.3).** Bump `helix_code` 1.11.0 → 1.12.0 to match helix_agent, or pin both to a single researched-latest.	Align both to the deep-web-researched latest gin v1.x (likely ≥1.12.0).	Touches both modules' build + middleware; needs the §11.4.99 latest-source check first.
OP-6	**Video-course + website production** (P2.4) — full production vs scripted-outline + asset stubs this cycle.	Author script/ storyboard + doc-graph assets; flag full video render as operator-gated.	Time/ resource budget; may need external tooling/ credentials.

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3. Phase / task structure (each task = RED → impl → GREEN+evidence → rollback)

Subagent-driven by default (§11.4.70/§11.4.103); ≥3 parallel non-contending streams. Phases ordered by dependency; P2.0 (D-1 bluff) lands FIRST as the highest-risk item (§11.4.132 risk-ordered).

P2.0 — D-1 bluff remediation: `/v1/completion/models` hardcoded list

Highest-risk, highest-visibility; a live CONST-036 / BLUFF-002 violation. Do first.

- **T2.0.1 RED — reproduce the bluff.** New test `internal/handlers/completion_models_test.go` with `RED_MODE` polarity switch (§11.4.115). `RED_MODE=1`: assert the response body is EXACTLY the 3 hardcoded ids (deepseek-coder / claude-3-sonnet-20240229 / gemini-pro) and that the set does NOT change when the registry/verifier is reconfigured → proves the list is static (the defect present on the pre-fix artifact). Capture output to `docs/qa/SP2-T2.0.1/`.
- **T2.0.2 IMPL — verifier-sourced list.** Rewrite `CompletionHandler.Models` (`internal/handlers/completion.go:411`) to return the live list from the verifier (`GetDiscoveredModels()` filtered `Verified==true`) joined with

`ProviderRegistry.ListProviders()` capabilities — never a literal map (CONST-036). Inject the registry/verifier into the handler (constructor wiring) rather than hardcode. Define cold/verifier-down behaviour: return the registry capabilities set with `verified:false` markers, never a fabricated "working" list (§11.4 PASS-bluff guard, mirrors SP1 P1.3.3).
- **T2.0.3 GREEN + evidence.** Flip `RED_MODE=0`: the same test becomes the standing regression guard (§11.4.135) — asserts the list now reflects registry/verifier state and CHANGES when a provider is enabled/disabled. Live integration run against a real verifier (real key where available) captured to `docs/qa/SP2-T2.0.2/`. Register guard `ATM-SP2-D1` in the standing suite.
- **Rollback:** revert `completion.go` to the literal-map version (single-commit, behind the test) if GREEN cannot be achieved; tracked item stays open. No data migration involved.

P2.1 — Unified catalog (the missing join layer)

- **T2.1.1 RED — absence test.** New test asserting there is currently NO single endpoint returning ensemble + helixllm + every provider + every verified model as uniformly-named entries (`RED_MODE=1` proves the 404 / absence). Evidence `docs/qa/SP2-T2.1.1/`.
- **T2.1.2 IMPL — catalog assembler.** New `internal/catalog/catalog.go`: `type Entry struct { Name string; Kind string /*ensemble|provider|model*/; Provider string; Verified bool; OverallScore float64; Enabled bool }` and `BuildCatalog(reg, verifier, ensembleSvc) []Entry`. Join rules: (a) one `ensemble` entry (+ presets per OP-3); (b) one `helixllm` entry + `helixllm/<model>` when enabled (OP from §P2.2.4); (c) one entry per `ProviderRegistry.ListProviders()`; (d) one `<provider>/<model_id>` entry per `DiscoveredModel` with `Verified==true` (NEVER static `SupportedModels`, NEVER hardcoded).
- **T2.1.3 IMPL — handler + route.** New `internal/handlers/catalog.go CatalogHandler.List`; register `GET /v1/catalog` (OP-1) inside the protected group near `internal/router/router.go:773` (alongside `/v1/providers`). Additive; no existing route changed.
- **T2.1.4 IMPL — selector resolution.** Extend the existing free-text `model` field (`CompletionRequest.Model`, `internal/handlers/completion.go:28`; ensemble closure `router.go:702`) so any catalog `Name` resolves to the right target: `ensemble[/preset]` → ensemble service; `helixllm[/model]` → helixllm provider; `<provider>[/<model>]` → that provider. Unknown selector → 400 with the candidate list (no silent guess, §11.4.6/§11.4.105).
- **T2.1.5 GREEN + evidence.** `RED_MODE=0` guard asserts the catalog lists all four classes and that each `Name` is selectable end-to-end (real proxied completion where a key exists). Liveness/captured evidence per §11.4.107 → `docs/qa/SP2-T2.1/`. Guard `ATM-SP2-CATALOG`.
- **Rollback:** new package + route are isolated; remove the route registration line + revert selector-resolution shim. Existing endpoints unaffected.

P2.2 — Per-provider individual exposure + HelixLLM promotion

- **T2.2.1 Provider catalog entry per discovered provider** (verifier-sourced) — covered by the catalog join (T2.1.2 clause c). Test: every `ListProviders()` name appears as a `kind:provider` entry with correct `enabled` flag.
- **T2.2.2 Per-provider working-model list** (`Verified==true`). Decision OP-2: default = derive from verifier join (no interface change). If operator chooses to add `GetModels(ctx)` to `LLMProvider (internal/llm/provider.go:9)`, that

becomes a SEPARATE sub-plan updating all 48 providers + their tests + the registry — RED-first per provider, evidence per provider.

- **T2.2.3 Naming scheme enforcement.** Test the canonical grammar against fixtures: `ensemble`, `ensemble/confidence_weighted`, `helixllm`, `helixllm/helixllm-default`, `anthropic/claude-3-sonnet-20240229`, `openrouter/x-ai/grok-4`, `groq`, `deepseek/deepseek-coder`. Truth-table test + a cell-flip mutation (§1.1) forcing FAIL (e.g. an UpperCase provider name must be rejected/normalised). Reuse existing lowercase names + the `vendor/model` slash form already present at `provider_registry.go:791` (`x-ai/grok-4`).
- **T2.2.4 Promote HelixLLM to first-class root.** Catalog emits a top-level `helixllm` entry (and `helixllm/<model>`) when `USE_HELIX_LLM=true` (`provider_registry.go:755`), NOT buried. Test: with the flag on, `helixllm` is a selectable root target; with it off, it is absent (no fabrication). Evidence `docs/qa/SP2-T2.2/`.
- **Rollback:** entries are produced by the join in T2.1.2; reverting that function reverts these.

P2.3 — SDK / API currency (clearly-bounded sub-plan)

Scope nuance (load-bearing): `submodules/helix_agent` has **NO vendor SDKs** — its 48 providers are hand-rolled `net/http`, so “update provider SDKs” there means **API-drift review of the hand-written clients vs each vendor’s CURRENT API reference**, not a `go.mod` bump. The cloud SDKs that CAN age out live in `helix_code/go.mod`. This sub-plan therefore spans both modules.

- **T2.3.1 Deep-web-research latest versions (§11.4.99).** For each flagged dep determine the TRUE latest via authoritative source (Go module proxy / vendor release notes), capture URLs + date in a `## Sources verified <date>` footer. Targets: `azcore` (v1.16.0), `azidentity` (v1.8.0 — auth/security, high priority), `aws-sdk-go-v2` core (v1.32.7), `aws-sdk-go-v2/config` (v1.28.7), `credentials` (v1.17.48), `bedrockruntime` (v1.23.1 — high value, new-model access), `getzep/zep-go/v3` (v3.10.0), `smacker/go-tree-sitter` (Aug-2024 pseudo-version — stale-likely; same pin in BOTH go.mods), `gin` (v1.11.0 helix_code / v1.12.0 helix_agent), `pgx/v5` (v5.7.6 / v5.9.2), `go-redis/v9` (v9.17.2 / v9.18.0). Output: `docs/research/SP2-sdk-currency/`.
- **T2.3.2 Hand-rolled provider API-drift audit.** For each of the 48 `internal/llm/providers/*`, diff the implemented request/response shape vs the vendor’s current API reference; open a tracked item per drift found (no silent assumption — §11.4.6). Prioritise providers with live keys for real-call verification.
- **T2.3.3 Bump cloud SDKs in `helix_code/go.mod` per OP-4 decision.** Per bump: RED (capture current build/behaviour) → `go get <dep>@<latest>` → `go mod tidy` → build + full unit/integration sweep → §11.4.92 multi-pass blast-radius → §11.4.108 four-layer (source/artifact/runtime/user-visible). Each bump is its OWN commit (revertible). Evidence: build logs + a real Bedrock/Azure call where credentials exist.
- **T2.3.4 Resolve gin skew (OP-5).** Align `helix_code` (v1.11.0) and `helix_agent` (v1.12.0) to the researched latest gin v1.x. Run BOTH modules’ route + middleware tests post-bump; capture green output. RED: a test asserting the two go.mods currently disagree (`RED_MODE=1`), GREEN: they agree post-fix.
- **Rollback:** every SDK bump is an isolated commit; `git revert` the specific bump + `go mod tidy`. No data migration. The hand-rolled-client audit produces tracked items only (no code change in this task).

P2.4 — Deliverables: docs / guides / graphs / SQL / website / video-course

All exported with `.html` + `.pdf` siblings via the Docs Chain engine (§11.4.106) + revision headers (§11.4.44).

- **T2.4.1 API reference + user guide** for `GET /v1/catalog` + the selector grammar + HelixLLM-as-root + per-provider/per-model targeting. Cross-reference vendor API docs verified in T2.3.1 (§11.4.99 footer).
- **T2.4.2 Architecture graphs** (mermaid/graphviz): the unified-catalog join (registry + verifier + ensemble → catalog → selector resolution); the four item-classes namespace tree.
- **T2.4.3 SQL** — if the catalog is persisted/cached, schema + sample queries for the catalog/verified-model tables; otherwise an SQL view modelling the catalog join for the workable-items / reporting DB. (Confirm persistence need with OP-1 design.)
- **T2.4.4 Website** — update `github_pages_website/` provider/model exposure pages to list the new namespace + examples (`ensemble`, `helixllm`, `<provider>/<model>`).
- **T2.4.5 Video-course** — script/storyboard for "selecting an exposure target" (OP-6 gates full render). Assets under `docs/qa/SP2-T2.4/`.

P2.5 — Full test-type matrix (cross-cutting, binds every task)

Per §11.4.27 / §11.4.50 / §11.4.85 / §11.4.98 / SP7. Each row = real infra (no mocks beyond unit), captured evidence, paired §1.1 mutation, fully automated (§11.4.98).

Test type	SP2 coverage
unit	catalog assembler join logic; selector grammar truth-table; Models-handler verifier-sourcing
integration	`GET /v1/catalog` against real `ProviderRegistry` + real verifier (real keys where available); selector → real completion
e2e	full user path: discover catalog → pick `ensemble` / `helixllm` / `<provider>/<model>` → real completion result
full-automation	self-driving, re-runnable `-count=3`, no human step (§11.4.98)
security	authz on `/v1/catalog` (behind `auth.Middleware`); no key/secret leakage in catalog output
ddos / scaling	catalog endpoint under load (delegated to HelixQA/Challenges per SP7 — evidence-backed, not config-only)
chaos / stress	verifier-down / registry-empty / partial-provider-failure → catalog degrades honestly (no fabricated "working" list); §11.4.85 injection
performance / benchmark	catalog assembly latency with N providers × M models
ui / ux	website exposure pages render the namespace correctly (CV/OCR per §11.4.117 if needed)

Test type	SP2 coverage
Challenges	`challenges/` bank entry exercising the unified catalog end-to-end
HelixQA	autonomous QA session over the new endpoints with captured wire evidence (SP7 P7.3)
regression-guards	`ATM-SP2-D1`, `ATM-SP2-CATALOG`, naming-grammar guard registered in standing suite (§11.4.135)

P2.6 — Code-review + release gating (§11.4.125 / §11.4.134)

- **T2.6.1** Dispatch code-review agent(s) over the full SP2 batch BEFORE pre-build + main build. Iterate until clean GO with ZERO findings AND ZERO warnings (§11.4.134), each round carrying captured physical evidence.
- **T2.6.2** Pre-build readiness verdict (§11.4.110) + clash detection (new handler ⇄ route registration ⇄ DI wiring). Coverage-completeness gate: every changed file → ≥1 gate + ≥1 test + ≥1 §1.1 mutation.
- **T2.6.3** Update `docs/CONTINUATION.md` + the session-resumption file (§11.4.131 / CONST-044) in the same commit as each state-advancing change.

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4. Files to create / modify

Path	Action	Purpose
`submodules/helix_agent/internal/handlers/completion.go` (:411 `Models`)	MODIFY	Replace hardcoded 3-model list (D-1) with verifier/registry-sourced list (CONST-036)
`submodules/helix_agent/internal/handlers/completion_models_test.go`	CREATE	RED→GREEN polarity test for D-1 (§11.4.115)
`submodules/helix_agent/internal/catalog/catalog.go`	CREATE	`Entry` type + `BuildCatalog()` join (ensemble+helixllm+providers+verified models)
`submodules/helix_agent/internal/catalog/catalog_test.go`	CREATE	Join logic + naming-grammar truth-table + §1.1 mutation
`submodules/helix_agent/internal/handlers/catalog.go`	CREATE	`CatalogHandler.List` for `GET /v1/catalog`
`submodules/helix_agent/internal/handlers/catalog_test.go`	CREATE	Handler integration test (real registry+verifier)
`submodules/helix_agent/internal/router/router.go` (near :773)	MODIFY	Register `GET /v1/catalog` (additive); wire selector resolution into ensemble/completion closures (:676/:702)

Path	Action	Purpose
`submodules/helix_agent/internal/handlers/completion.go` (:28 selector)	MODIFY	Selector resolution for catalog `Name` → target
`submodules/helix_agent/internal/services/provider_registry.go` (:750-764)	MODIFY (minimal)	Surface helixllm enabled-state for catalog promotion (read-only; no removal §11.4.122)
`submodules/helix_agent/internal/llm/provider.go` (:9)	MODIFY *(OP-2 only)*	Add `GetModels(ctx)` to interface — gated on OP-2; then 48 provider impls + tests
`helix_code/go.mod` / `go.sum`	MODIFY	SDK bumps (azidentity/azcore/bedrockruntime/aws-sdk-go-v2/zep/tree-sitter/pgx/go-redis) per OP-4; gin align per OP-5
`submodules/helix_agent/go.mod` / `go.sum`	MODIFY	gin align per OP-5; tree-sitter pin per T2.3.1
`docs/research/SP2-sdk-currency/*.md`	CREATE	§11.4.99 latest-source research + `Sources verified` footers
`docs/guides/SP2-exposure-catalog.md` (+ `.html`/`.pdf`)	CREATE	API reference + selector grammar + user guide
`docs/diagrams/SP2-catalog-join.*`	CREATE	Architecture graphs
`docs/sql/SP2-catalog-view.sql`	CREATE	Catalog view / cache schema (if persisted — confirm OP-1)
`github_pages_website/...`	MODIFY	Exposure-namespace pages
`challenges/...`	CREATE	End-to-end Challenge for the unified catalog
`docs/qa/SP2-*/`	CREATE	Captured runtime evidence per task
`docs/CONTINUATION.md` + session-resumption file	MODIFY	CONST-044 / §11.4.131 sync per state-advancing commit

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5. Ordered fine-grained task list

1. **T2.0.1** RED: reproduce D-1 hardcoded-list bluff (polarity `RED_MODE=1`), capture evidence.
2. **T2.0.2** IMPL: verifier/registry-source `CompletionHandler.Models`; define verifier-down honest behaviour.
3. **T2.0.3** GREEN: flip guard, live run, register `ATM-SP2-D1` in standing suite.
4. **T2.1.1** RED: catalog-absence test.
5. **T2.1.2** IMPL: `catalog.BuildCatalog()` join (4 item classes; `Verified==true` only).
6. **T2.1.3** IMPL: `CatalogHandler.List` + register `GET /v1/catalog` (OP-1).
7. **T2.1.4** IMPL: selector resolution for catalog `Name` into ensemble/helixllm/provider targets.
8. **T2.1.5** GREEN: end-to-end selectability, evidence, guard `ATM-SP2-CATALOG`.

9. **T2.2.1** Provider-entry-per-provider test.
10. **T2.2.4** HelixLLM first-class root promotion (flag-on/flag-off honesty test).
11. **T2.2.3** Naming-grammar truth-table + §1.1 mutation.
12. **T2.2.2** *(OP-2 gated)* per-provider working-model list; if interface change chosen, spawn the 48-provider `GetModels` sub-plan.
13. **T2.3.1** Deep-web-research latest SDK/API versions (§11.4.99); capture sources.
14. **T2.3.2** Hand-rolled 48-provider API-drift audit → tracked items.
15. **T2.3.4** Resolve gin 1.11.0⍰1.12.0 skew (OP-5), both modules' tests green.
16. **T2.3.3** Bump cloud SDKs in `helix_code/go.mod` per OP-4 (each bump = own commit + four-layer verify).
17. **T2.4.1–T2.4.5** Docs / graphs / SQL / website / video-course deliverables (OP-6 gates render).
18. **P2.5** Full test-type matrix execution (real infra, captured evidence, automated `-count=3`).
19. **T2.6.1** Code-review iterate-until-GO (§11.4.134).
20. **T2.6.2** Pre-build readiness verdict + clash + coverage-completeness gates.
21. **T2.6.3** CONTINUATION + resumption-file sync; commit/push merge-onto-latest-main (§11.4.113, no force).

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6. Rollback summary (per phase)

- **P2.0:** revert `completion.go:411` to literal-map (single commit behind the RED test).
- **P2.1/P2.2:** new isolated package `internal/catalog/` + one route registration + one selector shim — remove the route line + revert the shim; all existing endpoints untouched (additive design).
- **P2.3:** each SDK bump is its own revertible commit (`git revert` + `go mod tidy`); the API-drift audit emits tracked items only.
- **P2.4:** docs/website are additive; revert the doc commits.
- No destructive DB/data ops anywhere in SP2; no capability removed (§11.4.122).

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7. Anti-bluff / constitutional binding

- **CONST-036 / BLUFF-002:** D-1 fix + all catalog model sources are verifier/registry-derived; zero hardcoded model lists. "Working" = `DiscoveredModel.Verified==true` ONLY.
- **§11.4.115:** every fix RED-on-broken-artifact first with `RED_MODE` polarity switch.
- **§11.4.135:** every closed defect registers a permanent regression guard in the same commit.
- **§11.4.107:** PASS requires captured liveness evidence, never a single screenshot / metadata-only.
- **§11.4.99:** SDK/API currency uses latest-source web research with cited `Sources verified` footers.
- **§11.4.122:** HelixLLM is PROMOTED, never removed; all existing routes preserved.
- **§11.4.125/§11.4.134:** code-review iterate-until-GO before pre-build + main build.
- **§11.4.113:** merge-onto-latest-main; no force-push on any repo/submodule.
- **CONST-044 / §11.4.131:** CONTINUATION + resumption file synced each state-advancing commit.

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8. Summary (10 lines)

1. SP2 adds ONE unified-catalog root joining ensemble + HelixLLM + every provider + every VERIFIED model as uniformly-named selectable targets — the only missing layer; all building blocks exist.
2. Exposure surface confirmed = `submodules/helix_agent/internal/router/router.go` (Gin `/v1`); ensemble `:676`, providers `:773`, discovery `:1306-1313`, verification `:1378-1387`.
3. Highest-risk first: P2.0 fixes the D-1 hardcoded 3-model bluff at `internal/handlers/completion.go:411` (CONST-036/BLUFF-002), RED-first + standing guard.
4. New code is additive + isolated (`internal/catalog/` + `CatalogHandler` + `GET /v1/catalog` per OP-1); no existing route or capability removed (§11.4.122).
5. Naming scheme reuses existing lowercase provider names + the `vendor/model` slash form (`openrouter/x-ai/grok-4`); “working” = `DiscoveredModel.Verified==true`.
6. HelixLLM is promoted to a first-class root (`helixllm[/model]`) when `USE_HELIX_LLM=true`, flag-honest (no fabrication when off).
7. SDK currency is a bounded sub-plan: helix_agent has ZERO vendor SDKs (hand-rolled HTTP → API-drift audit), cloud SDKs bumped in `helix_code/go.mod`; gin 1.11.0⍰1.12.0 skew resolved (OP-5).
8. Six operator-input decisions flagged (OP-1 root path, OP-2 interface `GetModels`, OP-3 presets, OP-4 SDK blast radius, OP-5 gin direction, OP-6 video/website production).
9. Full deliverable set: docs/guides/graphs/SQL/website/video-course + the complete test-type matrix with captured evidence, paired §1.1 mutations, and fully-automated re-runnable tests.
10. Every closure is RED-first (§11.4.115) + regression-guarded (§11.4.135) + code-reviewed-until-GO (§11.4.134) + evidence-backed (§11.4.107); merge-onto-latest-main, no force-push (§11.4.113).

Task count: 21 ordered fine-grained tasks across 7 phases (P2.0–P2.6); 6 flagged operator-input decisions; ~18 file create/modify targets.